

HYDRAULIC SERVICES

PROPOSED OFFICE REFURBISHMENT
STANWELL POWER STATION
SWITCHYARD RD, STANWELL QLD
LOT 44 on SP140243

Rockhampton Regional Council

GENERAL NOTES

1. LAYOUTS OF SERVICES SHOWN ON THE ATTACHED DRAWINGS ARE DIAGRAMMATIC ONLY, THEY REPRESENT DESIGN INTENT ONLY IN ACCORDANCE WITH THE RELEVANT CODES AND STANDARDS, AND DOES NOT RELIEVE THE CONTRACTOR OR BUILDER FROM ANY DUE DILIGENCE IN RELATION TO BUILD ABILITY.
2. ALL INVERT LEVELS ON THE DRAWINGS ARE RECOMMENDED ONLY AND MUST BE CHECKED ON SITE BEFORE EXCAVATION OR INSTALLATION OF PIPEWORK TO ENSURE CONNECTION TO SUPPLY SERVICES AND CORRECT COVER AND FALL. ADVISE THE SUPERINTENDENT OF ANY APPARENT DISCREPANCIES BEFORE THE COMMENCEMENT OF ANY WORK; CLAIMS FOR REDUNDANT WORK AS A RESULT OF FAILING TO DO SO SHALL NOT BE CONSIDERED VALID.
3. ASCERTAIN THE DEPTH, POSITION AND SUITABILITY OF THE SANITARTY DRAIN CONNECTION POINTS PRIOR TO THE COMMENCEMENT OF ANY WORK (PIPELAYING & EXCAVATION) AND ENSURE THAT THE SANITARY DRAINAGE CAN GRAVITATE TO THE RELEVANT CONNECTION POINTS. THE SUPERINTENDENT SHALL BE ADVISED IMMEDIATELY SHOULD ANY ADJUSTMENT BE REQUIRED TO EXECUTE THE WORK. NO CLAIMS FOR REDUNDANT WORK WILL BE CONSIDERED DUE TO FAILURE TO COMPLY WITH THIS REQUIREMENT. NO PIPELAYING OR EXCAVATION SHALL BE UNDERTAKEN UNTIL THIS HAS BEEN UNDERTAKEN.
4. VISIT THE SITE BEFORE SUBMITTING FINAL TENDER TO ASSESS "ON SITE" CONDITIONS. FAILURE TO DO SO WILL FORFEIT ANY CLAIM FOR NOT BEING AWARE OF CONDITIONS AFFECTING THE TENDER.
5. NO WARRANTY IS GIVEN AS TO THE COMPLETENESS OR ACCURACY OF EXISTING SERVICES SHOWN ON THE ATTACHED DRAWINGS.
6. OBTAIN MEASUREMENTS AND OTHER NECESSARY INFORMATION AND COORDINATE THE DESIGN AND INSTALLATION IN CONJUNCTION WITH ALL TRADES BEFORE COMMENCING WORK.
7. ALL INVERT LEVELS SHOWN ON THE DRAWINGS ARE RECOMMENDATIONS ONLY AND MUST BE CHECKED ON SITE BEFORE EXCAVATION OR INSTALLATION OF PIPEWORK TO ENSURE CONNECTION TO SUPPLY SERVICES AND CORRECT COVER AND FALL CAN BE ACHIEVED. ADVISE THE CONSULTANT OF ANY APPARENT DISCREPANCIES BEFORE THE COMMENCEMENT OF ANY WORK; CLAIMS FOR REDUNDANT WORK AS A RESULT OF FAILING TO DO SO SHALL NOT BE CONSIDERED VALID.
8. THE CONTRACTOR SHALL ALLOW FOR ALL NECESSARY DIVERSIONS AND MINOR ADJUSTMENTS TO PIPEWORK AND EQUIPMENT AS MAY BE NECESSARY TO COMPLETE THE WORKS. ALL DIMENSIONING AND DETAILING NECESSARY FOR THE CORRECT PREPARATION OF WORKING DRAWINGS SHALL BE OBTAINED ON SITE AND NOT FROM THE ATTACHED DRAWINGS.
9. PRIOR TO COMMENCEMENT OF ANY ASSOCIATED WORKS THE CONTRACTOR SHALL CONFIRM THE PROXIMITY OF EXISTING UTILITY AND SITE SERVICES AND VISIT THE SITE TO BE FULLY AWARE OF SITE CONDITIONS. THE CONTRACTOR IS RESPONSIBLE FOR ANY COSTS ASSOCIATED WITH THE RECTIFICATION OF ANY DAMAGE TO EXISTING UTILITY AND SITE SERVICES, EXISTING STRUCTURES AND THE LIKE CAUSED BY CONTRACTOR NEGLIGENCE.
10. THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVIDING 'AS CONSTRUCTED DRAWINGS' FOR THE SUBMISSION TO THE LOCAL AUTHORITY AND AS DIRECTED BY THAT AUTHORITY. DRAWINGS MUST BE DRAFTED TO A STANDARD THAT IS THE SAME AS THE ORIGINAL
11. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO ENSURE THAT ALL MATERIALS USED HAVE WATER MARK OR PRODUCT APPROVAL
12. SANITARY FIXTURE POINTS SHOWN ON THESE DRAWINGS SHOULD NOT BE DIMENSIONED OR SCALED OFF PIPE SETOUT OR PENETRATIONS SHALL BE DIMENSIONED FROM ARCHITECTURAL DRAWINGS
13. BACKFLOW PREVENTION DEVICES SHALL BE INSTALLED WHERE REQUIRED BY THE PLUMBING CODE OF AUSTRALIA (NCC VOLUME 3), AS3500 AND THE LOCAL AUTHORITY HAVING JURISDICTION. DEVICES SHALL BE INSTALLED IN ACCORDANCE WITH AS3500
14. THE CONTRACTOR SHALL ALLOW FOR COST OF ALL FEES ASSOCIATED WITH THE PLUMBING WORKS SHOWN ON THESE DRAWINGS PAYABLE TO THE LOCAL AUTHORITY HAVING JURISDICTION
15. A SERVICE THAT PENETRATES A BUILDING ELEMENT THAT IS REQUIRED TO HAVE AN FRL MUST COMPLY WITH SECTIONS C4D13, C4D15 OF THE NATIONAL CONSTRUCTION CODE VOLUME 1 (BCA).
16. THE CONTRACTOR SHALL ENSURE AND VERIFY WITH BUILDING CERTIFIER THAT FIRE STOP COLLARS HAVE BEEN TESTED IN ACCORDANCE WITH AS1530.4 & AS4072.1 FOR THE PARTICULAR CIRCUMSTANCE OF INSTALLATION.
17. A SERVICE LABEL SHALL BE AFFIXED ADJACENT TO EACH SERVICE PENETRATION THAT REQUIRES A F.R.L AND THE LABEL SHALL PROVIDE INFORMATION IN ACCORDANCE WITH AS4072.1
18. THE PLUMBING CONTRACTOR SHALL LIAISE WITH STRUCTURAL, MECHANICAL AND ELECTRICAL CONTRACTOR'S BEFORE AND DURING CONSTRUCTION

SANITARY PLUMBING & DRAINAGE

1. ALL SANITARY DRAINAGE SHALL BE CONSTRUCTED IN AN APPROVED MATERIAL TO THE REQUIREMENTS OF THE RELEVANT CODES & AUTHORITIES. ALL PIPES SHOWN ARE 1000 HAVING A MINIMUM GRADE OF 1 IN 60 OR AS OTHERWISE SPECIFIED OR DIRECTED.
2. ALL PVC FITTINGS USED FOR SANITARY AND PREMISES SEWER SERVICES SHALL COMPLY WITH THE RELEVANT CODES & AUTHORITIES, STANDARDS & SHALL BE JOINTED EITHER BY RUBBER RINGS OR SOLVENT WELDING.
3. SUBCONTRACTOR SHALL ALLOW FOR ADDITIONAL EXCAVATION (INCLUDING ROCK) BACK FILL, PIPES, FITTINGS & ALL JUMP UPS TO THE LOCAL AUTHORITY REQUIREMENTS INCLUDING THOSE TO BRANCH DRAINS.
4. WHERE DRAINAGE LEVELS PERMIT, ALL BRANCH LINES SHALL BE GRADED &/OR LOWERED TO AVOID PENETRATING FOOTING BEAMS UNLESS SHOWN OTHERWISE. THE CONTRACTOR SHALL ALLOW FOR ADDITIONAL PIPE LENGTHS & WORK INCLUDING CONSTRUCTION OF ADDITIONAL INSPECTION OPENINGS IN THE SYSTEM AS REQUIRED BY LOCAL AUTHORITY.
5. PROVIDE INSPECTION OPENINGS IN THE SYSTEM AS REQUIRED BY AS3500.2.2021 SEC 4.7 OR AS REQUIRED BY THE RELEVANT AUTHORITY.
6. ALL VENTS TO TERMINATE WITH VENT COWL & ABLE TO ACHIEVE SEPARATION DISTANCES IN ACCORDANCE WITH AS3500.2.2021 FIGURE 6.9.5. ROOF TO BE LEFT WATER TIGHT.

EQUIVALENT PIPE DIAMETERS (DN)					
DN	COPPER (TYPE B)	PE (SDR 11)	PE-Xa	PB (CLASS 16)	PP-R (SDR 9)
15	15 (10.88mm ID)	16 (12.7mm ID)	16 (10.8mm ID)	18 (12.5mm ID)	-
18	18 (13.84mm ID)	20 (16mm ID)	20 (13.6mm ID)	22 (17.6mm ID)	-
20	20 (17.01mm ID)	25 (20.2mm ID)	25 (18.0mm ID)	28 (22.8mm ID)	-
25	25 (22.96mm ID)	32 (26mm ID)	32 (23.2mm ID)	-	32 (24.87mm ID)
32	32 (29.31mm ID)	40 (32mm ID)	40 (29.9mm ID)	-	40 (31mm ID)
40	40 (35.66mm ID)	50 (40.4mm ID)	50 (36.2mm ID)	-	50 (38.8mm ID)

WATER SERVICE NOTES

1. INSTALL PIPEWORK IN STRAIGHT LINES & UNIFORM GRADES. ARRANGE & SUPPORT THE PIPEWORK SO THAT IT REMAINS FREE FROM VIBRATION WHILST PERMITTING THERMAL MOVEMENT. KEEP THE NUMBER OF JOINTS TO A MINIMUM. PREVENT DIRECT CONTACT BETWEEN INCOMPATIBLE METALS.
2. PROVIDE COMPLIANCE WITH THE NATIONAL CONSTRUCTION CODE 2022 VOLUME 3 SECTION B2D5 AND B2D6 IN RESPECT TO TEMPERATURES OF HEATED DELIVERY OF WATER TO SANITARY FIXTURES.
3. STAMPED & TESTED ISOLATING OR LOOSE VALVE STOPCOCKS SHALL BE SUPPLIED & INSTALLED GENERALLY WHERE SHOWN ON THE DRAWINGS & WHEREVER NECESSARY TO PROVIDE CONTROL OF INDIVIDUAL BRANCHES AND/OR GROUPS OF FIXTURES. WHERE APPROPRIATE, RECESS COCKS SHALL BE PROVIDED AS ISOLATORS.
4. ALL TESTS REQUIRED BY THE AUTHORITIES SHALL BE CARRIED OUT BY THE CONTRACTOR. SUPPLY ALL PLANT & MATERIAL TO CARRY OUT SUCH TESTS & RE-TESTING IF REQUIRED.
5. PROVISION SHALL BE MADE TO ENSURE THAT THE MAXIMUM STATIC PRESSURE AT ANY OUTLET, OTHER THAN A FIRE SERVICE OUTLET, WITHIN A BUILDING DOES NOT EXCEED 500 KPA.
6. NOMINAL PIPE DIAMETERS (DN) SHOWN ON THESE DRAWINGS ARE FOR TYPE B COPPER (U.N.O.). WHERE AN ALTERNATIVE MATERIAL IS PROPOSED, A PIPE MATERIAL WITH AN INTERNAL PIPE DIAMETER EQUIVALENT OR LARGER TO TYPE B COPPER SHALL BE TAKEN FROM AS3500.1 2021 APPENDIX A
7. COLD WATER PIPES INSTALLED INGROUND BENEATH CONCRETE SLAB SHALL GENERALLY COMPLY WITH AS3500.1 2021 CLAUSE 5.5.4. PIPE UNDER SLAB FLOOR SHALL BE CONTINUOUS WITHOUT JOINTS. METAL PIPES SHALL BE LAGGED WITH A IMPERMEABLE MATERIAL.
8. HOT WATER PIPES INSTALLED INGROUND BENEATH CONCRETE SLAB SHALL GENERALLY COMPLY WITH AS3500.4 CLAUSE 4.6.1.3. PIPE UNDER SLAB FLOOR SHALL BE CONTINUOUS WITHOUT JOINTS AND INSULATED IN ACCORDANCE WITH AS3500.4 2021 CLAUSE 8.2.
9. FLOW RESTRICTION DEVICES SHALL BE FITTED WHERE TAP FLOW RATES ARE NOT SELF LIMITING DEVICES SHALL BE USED TO RESTRICT FLOWS TO BASINS, SINKS, LAUNDRY TROUGHS AND SHOWERS TO A MAXIMUM FLOW RATE OF 9L/M OR AS DIRECTED BY THE AUTHORITY HAVING JURISDICTION.
10. TAP POINTS SHOWN ON THESE DRAWINGS ARE INDICATIVE ONLY AND DO NOT NECESSARILY REFLECT THE EXACT LOCATION OF SUPPLY TO THE FIXTURE. REFER TO ARCHITECTURAL DRAWINGS AND/OR END CLIENT PC SPECIFICATIONS FOR INSTALLATION REQUIREMENTS AND TO ESTABLISH SUPPLY PROVISIONS
11. COLD WATER PIPEWORK SHALL BE IDENTIFIED IN ACCORDANCE WITH AS1345 AND AS3500.1 2021 CLAUSE 5.19
12. HOT WATER PIPEWORK SHALL BE IDENTIFIED IN ACCORDANCE WITH AS1345 AND AS3500.4 2021 CLAUSE 4.13.2
13. PIPES SHALL BE SEPARATED FROM OTHER SERVICES IN ACCORDANCE WITH AS3500.1 2021 CLAUSE 5.3
14. COLD WATER PIPE SYSTEM SHALL BE TESTED AND COMMISSIONED IN ACCORDANCE WITH AS3500.1 2021 SECTION 17
15. HOT WATER PIPE SYSTEM SHALL BE TESTED AND COMMISSIONED IN ACCORDANCE WITH AS3500.4 2021 SECTION 9
16. HOT WATER PIPE INSULATION SHALL FULLY COMPLY WITH AS3500.4 2021 SECTION 8
17. PIPE SIZE LIMITATIONS SHALL BE IN ACCORDANCE WITH AS3500.1.2021 CLAUSE 3.5
18. WATER SAVING SHOWERS, BASINS, SINKS & LAUNDRY TROUGHS WITH AAA RATING OR 3 STAR 'WELS' SHALL BE INSTALLED
19. MAXIMUM 6-3 (WELS 4 STAR) DUAL FLUSH WC SHALL BE INSTALLED

MATERIALS - SANITARY PLUMBING, DRAINAGE & VENTS			
ABOVE GROUND			
LINETYPE		SIZE	MATERIAL
 SAN	SANITARY PLUMBING	DN40-DN80	DWV PIPE & FITTINGS SH, SOLVENT WELD JOINT
 SAN	SANITARY PLUMBING	DN100	DWV PIPE & FITTINGS SN6 OR SN10, SOLVENT WELD JOINT
	VENT PIPE	AS NOTED	DWV PIPE & FITTINGS SH, SOLVENT WELD JOINT DWV PIPE & FITTINGS SN6 OR SN10, SOLVENT WELD JOINT
BELOW GROUND			
 SAN	SANITARY DRAINAGE	DN40-DN80	DWV PIPE & FITTINGS SH, SOLVENT WELD JOINT
 SAN	SANITARY DRAINAGE	DN100	DWV PIPE & FITTINGS SN6 OR SN10, SOLVENT WELD JOINT
	VENT PIPE	AS NOTED	DWV PIPE & FITTINGS SH, SOLVENT WELD JOINT DWV PIPE & FITTINGS SN6 OR SN10, SOLVENT WELD JOINT

MATERIALS - WATER SERVICES			
ABOVE GROUND - COLD			
LINETYPE		SIZE	MATERIAL
	POTABLE WATER	DN16-DN25	TYPE B COPPER WITH SILVER BRAZED OR APPROVED CRIMPED JOINTS
	POTABLE WATER	DN16-DN32	REHAU RAUTITAN PE-XA - PLATINUM

BELOW GROUND - COLD			
	POTABLE WATER	DN16-DN32	PE100 SERIES PRESSURE PIPE (BLUELINE) CLASS 12.5, COMPRESSION JOINTS
	POTABLE WATER	DN16-DN25	TYPE B COPPER WITH SILVER BRAZED OR APPROVED CRIMPED JOINTS
	POTABLE WATER	DN16-DN32	REHAU RAUTITAN PE-XA - PLATINUM

ABOVE GROUND - HOT			
	POTABLE WATER (60°)	DN16-DN20	TYPE B COPPER WITH SILVER BRAZED OR APPROVED CRIMPED JOINTS
	POTABLE WATER (60°)	DN16-DN25	REHAU RAUTITAN PE-XA - PLATINUM
	POTABLE WATER (45°-50°)	DN16-DN25	REHAU RAUTITAN PE-XA - PLATINUM

BELOW GROUND - HOT			
	POTABLE WATER (60°)	DN16-DN20	TYPE B COPPER WITH SILVER BRAZED OR APPROVED CRIMPED JOINTS
	POTABLE WATER (60°)	DN16-DN25	REHAU RAUTITAN PE-XA - PLATINUM
	POTABLE WATER (45°-50°)	DN16-DN25	REHAU RAUTITAN PE-XA - PLATINUM

PE-XA RULES

WITHIN CONCRETE - *Fittings shall be wrapped with suitable tape to avoid direct contact with concrete. Pipe must be laid in conduit.*

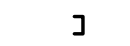
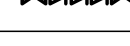
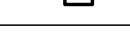
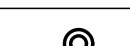
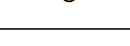
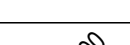
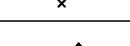
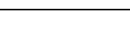
BELOW GROUND - *All joints in ground, except for RAUTITAN PX / PXsr fitting joints with RAUTITAN PE-Xa pipes, must be protected against corrosion using a recognized corrosion protection system, such as the Denso DEKOTEC N15/PE5 system or equivalent.*

UNDER SLAB - AS 2870-2011 "Residential slabs and footings" requires cold water pipes and heated or hot water pipes installed under a slab to be installed within a conduit so that if the pipe leaks water it will be noticed above the slab or outside the slab and will not leak unnoticed under the slab.

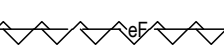
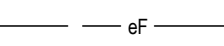
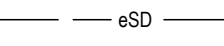
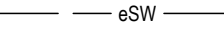
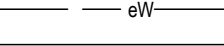
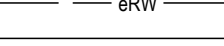
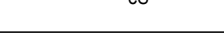
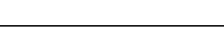
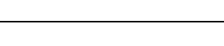
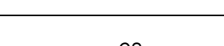
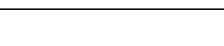


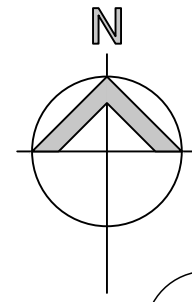
1 Locality Plan

NOT TO SCALE

LEGEND	
	AIR ADMITTANCE VALVE
	CAPPED END
	BALL VALVE
	SINGLE CHECK VALVE
	DUAL CHECK VALVE
	DOUBLE CHECK VALVE (TESTABLE)
	REDUCED PRESSURE ZONE DEVICE (TESTABLE)
	OVERFLOW RELIEF GULLY/DISCONNECTOR GULLY
	END LINE
	OUTLET
	EXPANSION JOINT
	FLOOR WASTE GULLY
	ISOLATION VALVE IN PATH BOX
	PIPE DROP
	S AND/OR P TRAP
	PIPE DROP
	TEMPERING VALVE (50° MAX.)
	THERMOSTATIC MIXING VALVE (45° MAX)
	TAP POINTS (HOT)
	TAP POINTS (COLD)
	TAP POINTS (TEPID)
	PROPOSED LEVEL
	EXISTING LEVEL
	RISER SERVICE TYPE
	SIZE
	ITEM NUMBER

ABBREVIATIONS	
AAV	AIR ADMITTANCE VALVE
BN	BASIN
BWU	BOILING WATER UNIT
CS	CLEANERS SINK
CW	COLD WATER
DCV	DOUBLE CHECK VALVE
DF	DRINKING FOUNTAIN
DWM	DISH WASHING MACHINE
FFL	FINISHED FLOOR LEVEL
FHR	FIRE HOSE REEL
FWG	FLOOR WASTE GULLY
HWU	HOT WATER UNIT
HW	HOT WATER
IL	INVERT LEVEL
IO	INSPECTION OPENING
IOS	INSPECTION OPENING TO SURFACE
ORG	OVERFLOW RELIEF GULLY
RL	RELATIVE LEVEL
SK	SINK
SHR	SHOWER
STY	SAFETRAY
TD	TUNDISH
TMV	THERMOSTATIC MIXING VALVE
TPV	TEMPERING VALVE
TTD	TRAPPED TUNDISH
TR	TROUGH
UR	URINAL
WC	WATER CLOSET

LINETYPES	
	REDUNDANT SERVICE
	EXISTING FIRE SERVICE
	EXISTING SANITARY DRAIN
	EXISTING STORMWATER
	EXISTING WATER SERVICE
	EXISTING RAW WATER
	EXISTING SEWER
	EXISTING ELECTRICAL MAIN
	EXISTING TELECOMMUNICATION
	EXISTING TRADE WASTE
	CIVIL STORMWATER



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TENDER ONLY

NOT FOR
CONSTRUCTION

DRAWING SHOULD BE
READ IN COLOUR PRINT

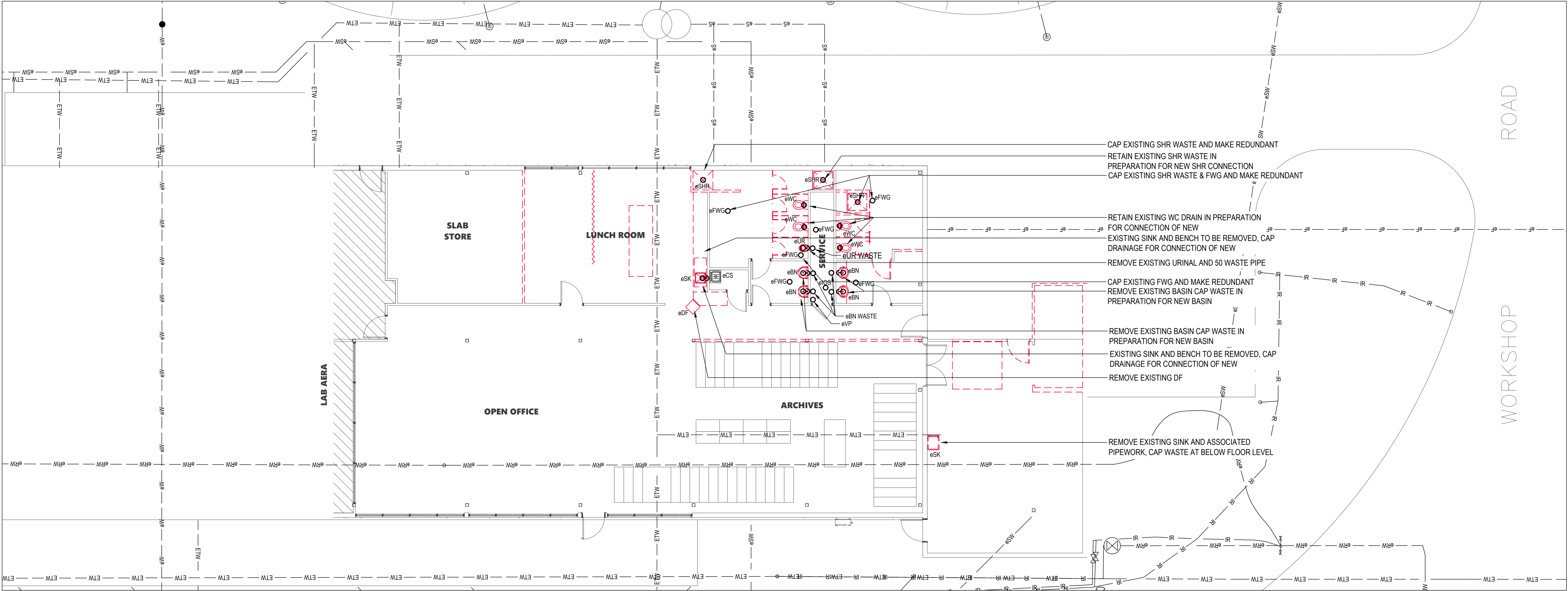
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P2	ISSUED FOR COORDINATION	30.01.2026
P1	ISSUED FOR REVIEW	10.12.2025
ISSUE:	DESCRIPTION:	DATE:
 HYDRO-TECH DESIGNS PTY LTD HYDRAULIC AND FIRE PROTECTION CONSULTANTS Tel: 0438 433 544 Email: rch@bigpond.net.au		



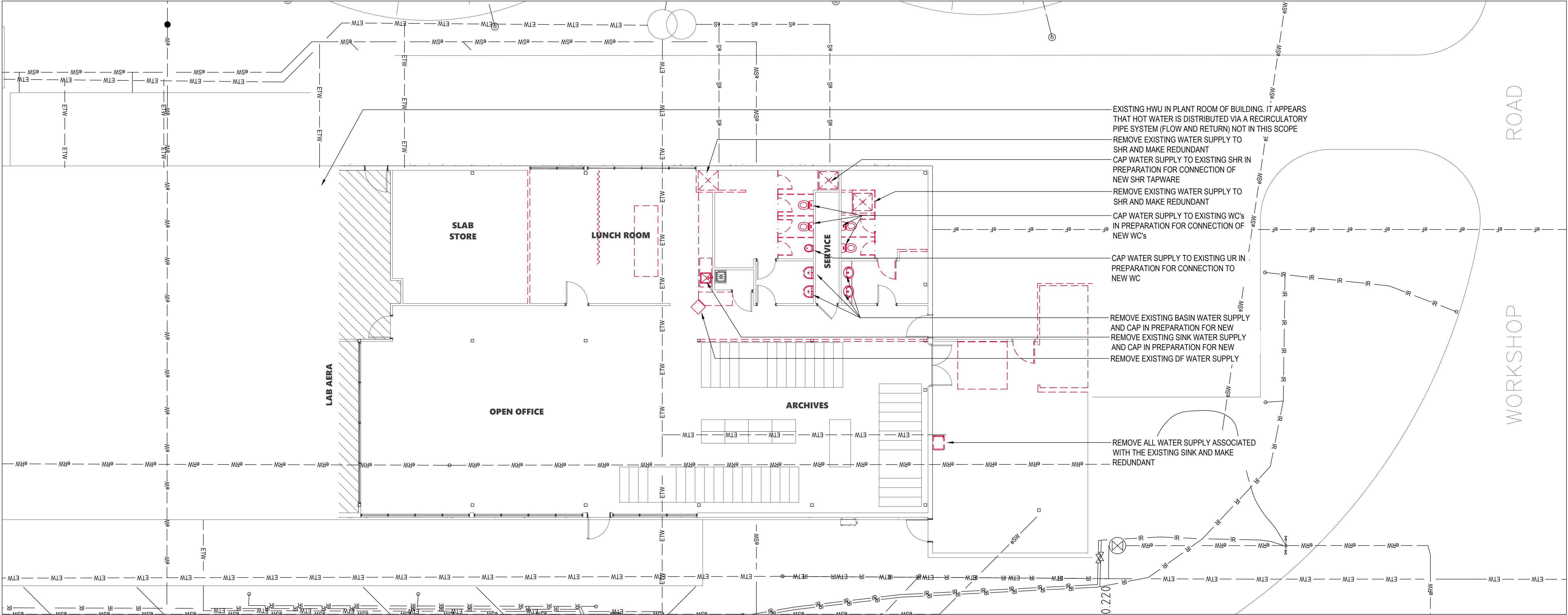
PROJECT:
REGISTRY BUILDING
OFFICE REFURBISHMENTS
STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL

TITLE:
HYDRAULIC SERVICES
COVER SHEET
NOTES
LOCALITY PLAN

Datum:	Status:	
TENDER		
Scale:	Date:	
-	19.11.2025	
Project Number.	Sheet	Issue.
HS-0690-c	H.01.01	01



1 Existing Floor Plan - Drainage demolition
Scale: 1:100 @ A1



2 Existing Floor Plan - Water demolition
Scale: 1:100 @ A1

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01 ISSUED FOR TENDER 24.02.2026
P2 ISSUED FOR COORDINATION 30.01.2026
P1 ISSUED FOR REVIEW 10.12.2025
ISSUE: DESCRIPTION: DATE:
HTD HYDRO-TECH DESIGNS PTY LTD
HYDRAULIC AND FIRE PROTECTION CONSULTANTS
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CLIENT:
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BUILDING DESIGNERS

BUILDING DESIGNER
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BUILDING DESIGNERS

PROJECT:
REGISTRY BUILDING
REFURBISHMENTS
STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL
TITLE:
HYDRAULIC SERVICES
DEMOLITION PLAN
DRAINAGE AND WATER

Datum: Status:
TENDER
Scale: Date:
AS NOTED 19.11.2025
Project Number. Sheet Issue.
HS-0690-c H.02.01 01

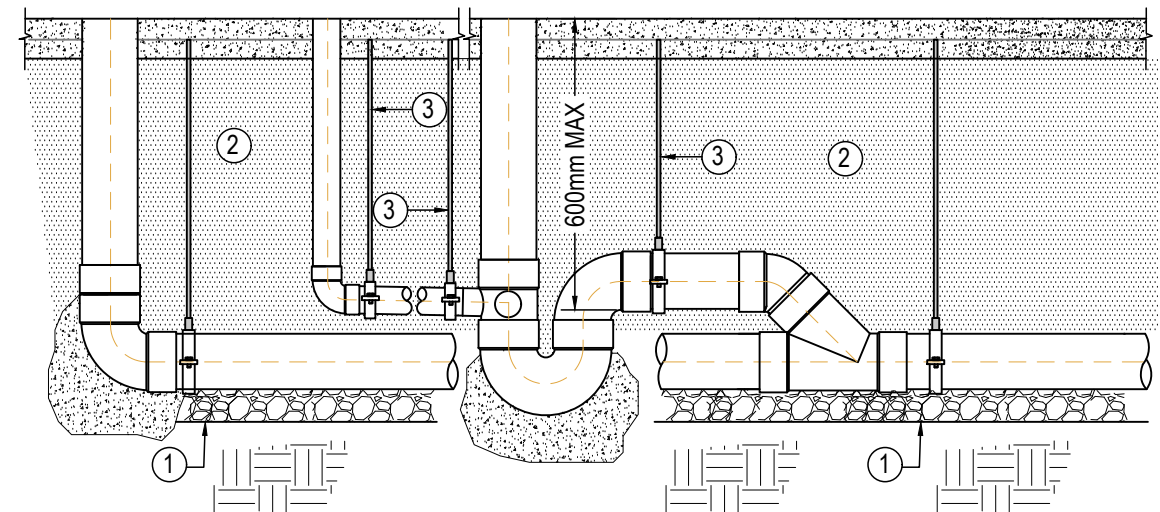
NOTE: A SITE CLASSIFICATION REPORT IS NOT AVAILABLE FOR THIS SITE. IN ACCORDANCE WITH BUILDING AND PLUMBING NEWSFLASH 569 A PIPE ARTICULATION DESIGN WILL BE BASED ON MAXIMUM POTENTIAL MOVEMENT THAT COULD BE EXPECTED ON SITE, TYPE 'E' WILL BE USED FOR THE DESIGN.

NEWSFLASH 569 EXTRACT-
IN ORDER TO ENSURE THAT WORK CAN CONTINUE, A LOCAL GOVERNMENT MAY ACCEPT A PERMIT APPLICATION WITHOUT A SOIL CLASSIFICATION REPORT IF:

- AN ARTICULATION DESIGN IS PROVIDED BASED ON THE MAXIMUM POTENTIAL MOVEMENT THAT COULD BE EXPECTED FOR THE SITE. FOR EXAMPLE, WHERE THE WORK INVOLVES RENOVATION OF A SHOP FIT OUT IN AN EXISTING COMMERCIAL PREMISES LOCAL GOVERNMENT COULD ACCEPT AN ARTICULATION DESIGN BASED ON AN 'E' SOIL CLASSIFICATION SITE WITH EXPECTED MOVEMENT OF 150 MILLIMETRES

NOTES

- THE FOLLOWING NOTES AND DETAILS PROVIDED ON THIS SHEET ARE A GUIDE ONLY FOR SANITARY PLUMBING, DRAINAGE AND SHOULD BE READ IN CONJUNCTION WITH AS/NZS3500, AS 2870 AND ANY OTHER RELEVANT STANDARD.
- ALL DRAINS INSTALLED IN GROUND THAT IS FILLED, UNSTABLE, CONTAINS EXPANSIVE CLAYS (M, H, E AND P SITES) OR WATER CHARGED AND WHERE SOIL MOVEMENT MAY EFFECT THE PERFORMANCE OF ANY DRAIN, A PLAN AND SPECIFICATION TOGETHER WITH A SOIL REPORT AND CALCULATIONS SHOWING METHODS TO PROTECT THE DRAINAGE SHALL BE LODGED WITH THE LOCAL AUTHORITY INCLUDING THE REQUIREMENTS OF THE COMPLIANCE PERMIT AND ASSESSMENT.
- ALL PLUMBING TO BE CONSTRUCTED IN ACCORDANCE WITH AS/NZS 3500 AND THE REQUIREMENTS OF AS 2870 SECTION 5 - CLAUSE 5.5 AND SECTION 6: CLAUSE 6.6. ARTICULATION AND EXPANSION JOINTS ARE TO BE PROVIDED TO ACCOMMODATE MOVEMENT IN ALL PIPE WORK WITHIN 3 METRES OF THE DWELLING AND COMPLY WITH AS1260.
- GRADES IN PIPE WORK ON 'E' SITES SHALL HAVE MINIMUM GRADE OF 1 IN 50 (U.N.O.)
- ALL PIPE WORK PASSING THROUGH CONCRETE MUST HAVE 25mm MIN. LAGGING.



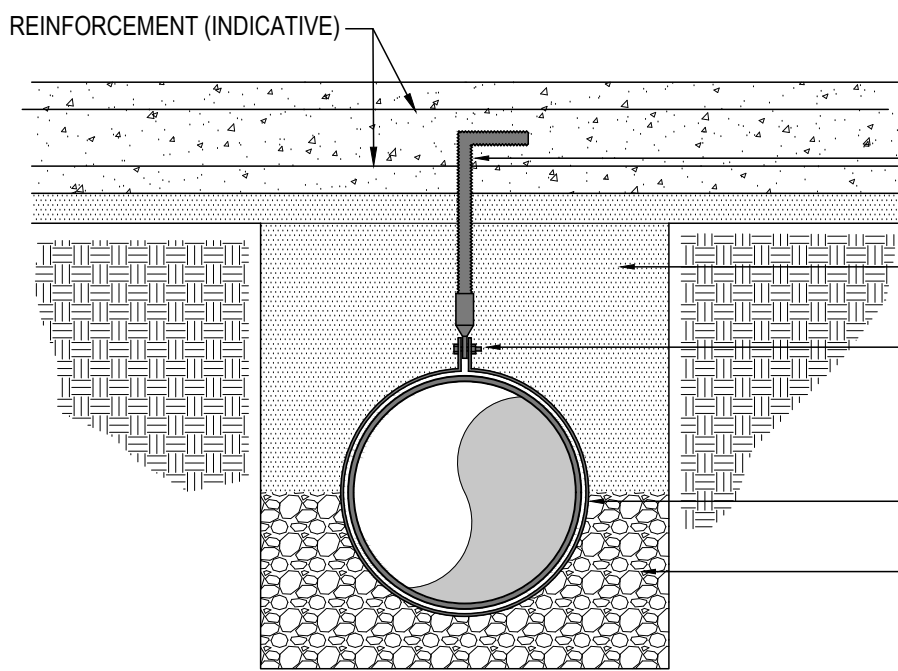
RECOMMENDED PIPE SUPPORT LOCATIONS ON "E" TYPE SOILS
NOT TO SCALE

NUMBERED NOTES

- DRAINAGE GRAVEL
- BACKFILL TRENCH WITH SAND
- SUPPORT BRACKET SPACED AT MAX 1.2m ALONG MAIN DRAINS AND TYPICALLY WHERE SHOWN ON DETAIL TO SUPPORT PIPE RISERS AND TRAPS. REFER TO DETAIL 'A'

SOIL CLASSIFICATION TABLE

AS2870 CLASSIFICATION	DIFFERENTIAL MOVEMENT	RECOMMENDED MINIMUM GRADE	SOIL FOUNDATION
E	76mm+	1:50	EXTREMELY REACTIVE SITES, WHICH MAY EXPERIENCE EXTREME GROUND MOVEMENT FROM MOISTURE CHANGES



DETAIL - 'A': SUPPORT BRACKET FOR DRAINS
BENEATH BUILDING FLOOR
NOT TO SCALE

SANITARY DRAIN SCHEDULE

FIXTURE TYPE	SIZE
SINK TO DRAIN	50 TRAPPED
WC TO DRAIN	100
BASIN TO DRAIN	40 TRAPPED
TROUGH TO DRAIN	50 TRAPPED
URINAL TO DRAIN	40 TRAPPED, PIPE IN WALL
BASIN TO FWG	40 TRAPPED, PIPE IN WALL
TD TO FWG	40
SHR TO FWG	50 (80 or 100 PALAZZI TRAP IF >2.5 FROM FWG)
SHR TO DRAIN	100x65 PALAZZI GULLY

MINIMUM PIPE GRADES (SANITARY DRAINAGE)

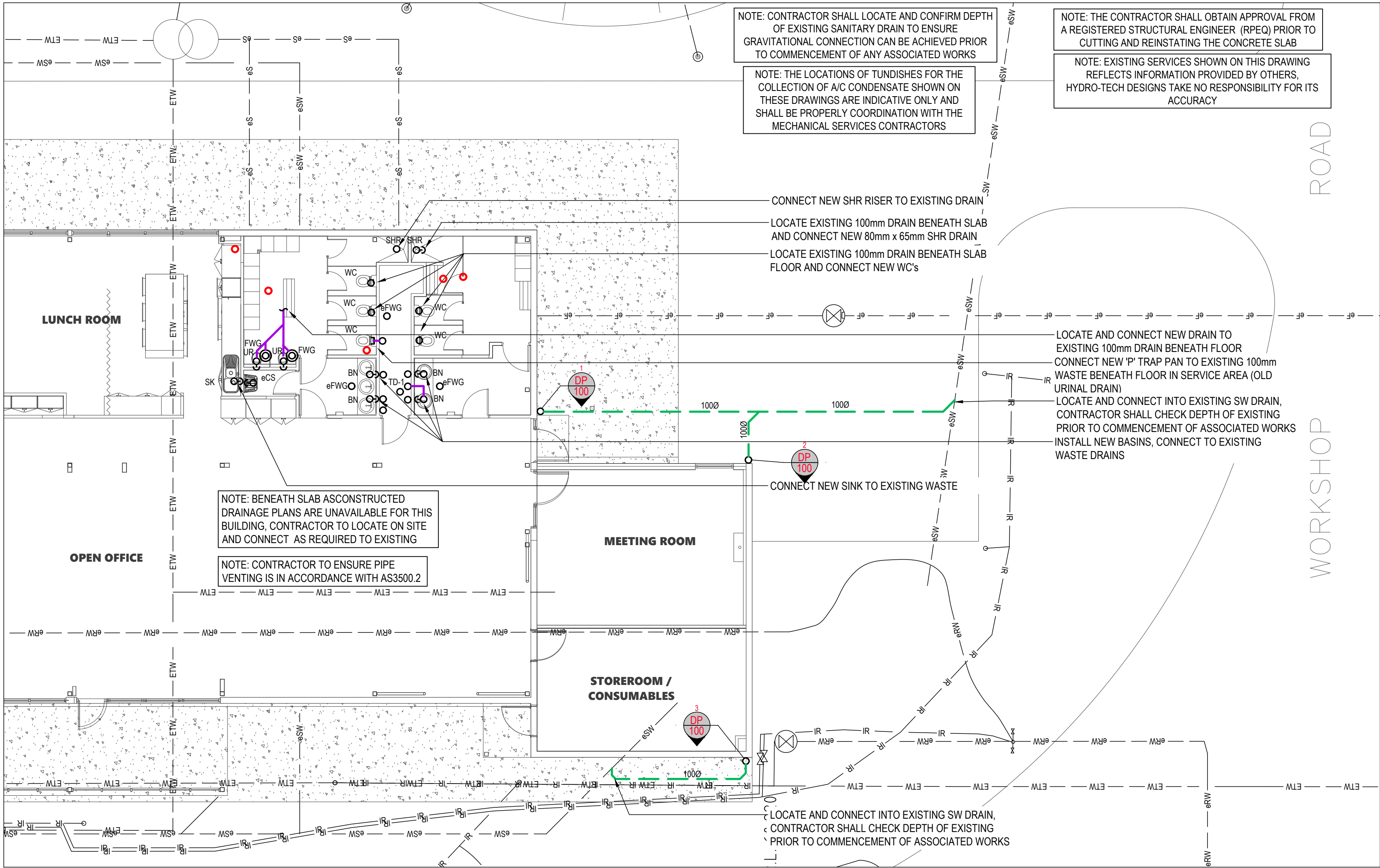
Nominal Size (DN)	Minimum Grade (%)
40-65	2.50
80	1.65
100	1.65*

LEGEND: WATER

BWU-1	UNDERBENCH CHILLED/BOILING WATER UNIT EQUAL TO ZIP HYDROTAP G5 BC40 CLASSIC 1.4kW 220-240V 50/60Hz 1 x 10A GPO, INSTALL TO MANUFACTURERS Zip
HC-1	SHROUDED C/P HOSE TAP ON AT LOW LEVEL ADJACENT TO URINAL SPECIFICALLY TO MAINTAIN WATER SEAL IN FWG

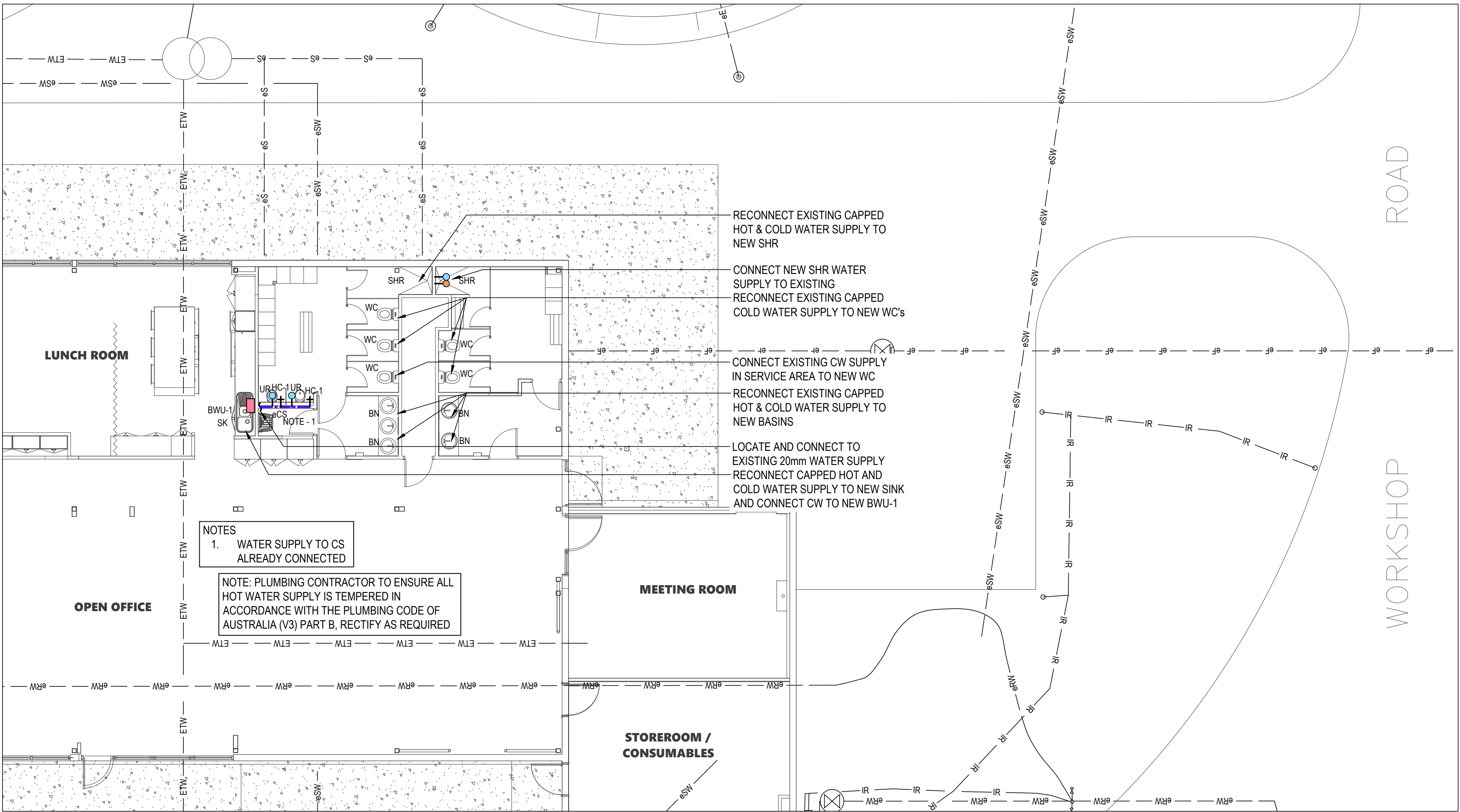
LEGEND: DRAINAGE

TD-1	100 x 40 TUNDISH CONNECTED TO BASIN FOR RECEIVING A/C CONDENSATE (REFER TO DETAIL)
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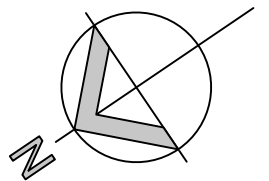
1 Proposed Floor Plan - Drainage

Scale: 1:100 @ A1



2 Proposed Floor Plan - Water

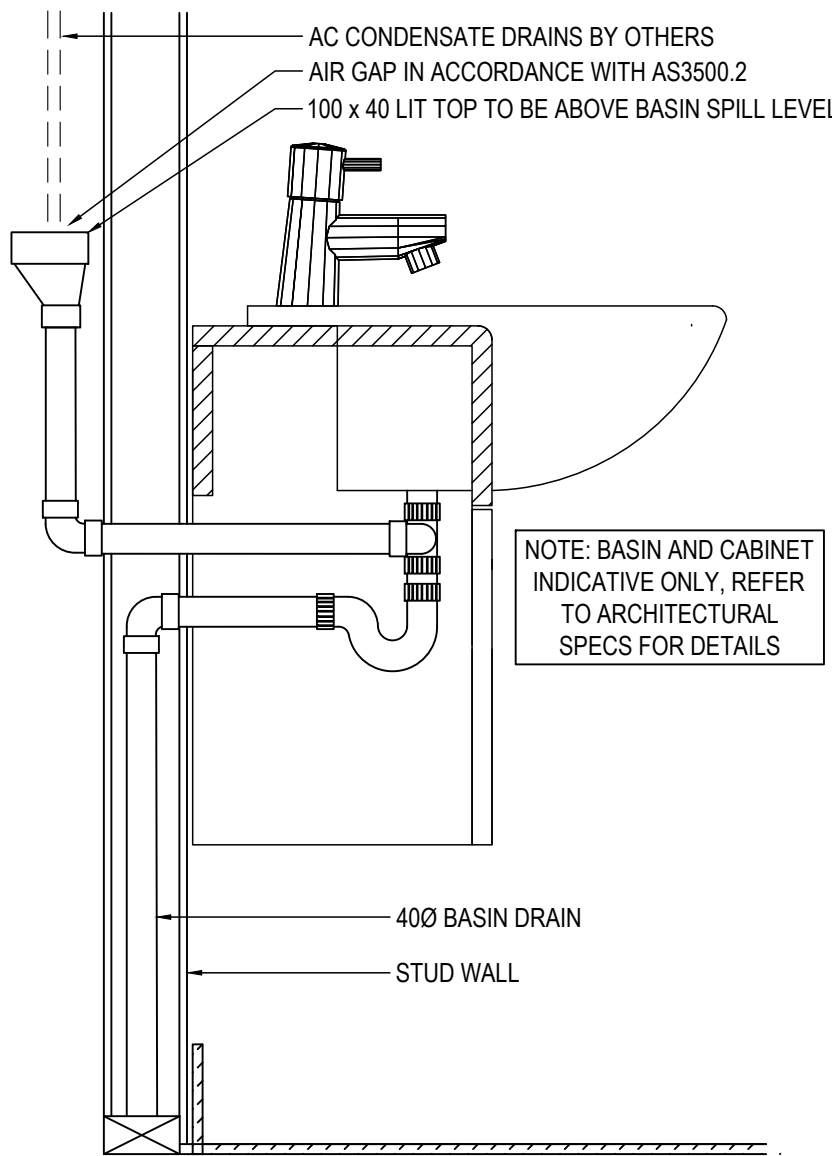
Scale: 1:100 @ A1



Issued for
TENDER ONLY

NOT FOR
CONSTRUCTION

DRAWING SHOULD BE
READ IN COLOUR PRINT



A/C WASTE WATER TUNDISH DETAIL (TD-1)
NOT TO SCALE

02	AMENDMENTS	06.03.2026
01	ISSUED FOR TENDER	24.02.2026
P2	ISSUED FOR COORDINATION	30.01.2026
P1	ISSUED FOR REVIEW	10.12.2025

ISSUE: DESCRIPTION: DATE:

HTD HYDRO-TECH DESIGNS PTY LTD
HYDRAULIC AND FIRE PROTECTION CONSULTANTS
Tel: 0438 433 544
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CLIENT:



BUILDING DESIGNER



PROJECT:
REGISTRY BUILDING
REFURBISHMENTS
STANWELL POWER STATION
SWITCHYARD ROAD, STANWELL

TITLE:
HYDRAULIC SERVICES
PROPOSED GROUND FLOOR PLANS
DRAINAGE & WATER

Datum: Status:
TENDER

Scale: Date:
AS NOTED 19.11.2025

Project Number. Sheet Issue.
HS-0690-c H.03.01 02